

Calculus I Lesson 20

$$1) \tan x = \frac{\sin x}{\cos x}$$

$$a) D_x(\sin x) = \underline{\quad ? \quad}$$

$$b) D_x(\cos x) = \underline{\quad ? \quad}$$

$$c) D_x(\tan x) = \frac{\cos x \cdot D_x(\sin x) - \sin x \cdot D_x(\cos x)}{[\cos x]^2} = \underline{\quad ? \quad} \quad \text{Using Quotient Rule}$$

$$2) \cot x = \frac{\cos x}{\sin x}$$

$$a) D_x(\cos x) = \underline{\quad ? \quad}$$

$$b) D_x(\sin x) = \underline{\quad ? \quad}$$

$$c) D_x(\cot x) = \frac{\sin x \cdot D_x(\cos x) - \cos x \cdot D_x(\sin x)}{[\sin x]^2} = \underline{\quad ? \quad}$$

Using Quotient Rule

$$3) \sec x = \frac{1}{\cos x}$$

$$a) D_x(1) = \underline{\quad? \quad}$$

$$b) D_x(\cos x) = \underline{\quad? \quad}$$

$$c) D_x(\sec x) = \frac{\cos x \cdot D_x(1) - 1 \cdot D_x(\cos x)}{[\cos x]^2} = \underline{\quad? \quad}$$

Using Quotient Rule

$$4) \csc x = \frac{1}{\sin x}$$

$$a) D_x(1) = \underline{\hspace{1cm}} ?$$

$$b) D_x(\sin x) = \underline{\hspace{1cm}} ?$$

$$c) D_x(\csc x) = \frac{\sin x \cdot D_x(1) - 1 \cdot D_x(\sin x)}{[\sin x]^2} = \underline{\hspace{1cm}} ?$$

Using Quotient Rule

5) $h(x) = 2x^3 - \tan x + \cos x$

$h'(x) = \underline{\hspace{2cm}}?$

6) $h(x) = 2x^{3/2} - \cot x + \sin x$

$h'(x) = \underline{\hspace{2cm}}?$

$$7) \ h(x) = \frac{\sin^2 x}{\cos x}$$

$$\text{Hint: } \frac{\sin^2 x}{\cos x} = \frac{1 - \cos^2 x}{\cos x} = \frac{1}{\cos x} - \frac{\cos^2 x}{\cos x} = \sec x - \cos x$$

$$h'(x) = \underline{\hspace{2cm}} ?$$

$$8) \ h(x) = \frac{\sec^2 x}{\tan x}$$

$$\text{Hint: } \frac{\sec^2 x}{\tan x} = \frac{1 + \tan^2 x}{\tan x} = \frac{1}{\tan x} + \frac{\tan^2 x}{\tan x} = \cot x + \tan x$$

$$h'(x) = \underline{\hspace{1cm} ? \hspace{1cm}}$$

$$9) \ h(x) = \frac{\csc^2 x}{\cot x}$$

$$\text{Hint: } \frac{\csc^2 x}{\cot x} = \frac{1 + \cot^2 x}{\cot x} = \frac{1}{\cot x} + \frac{\cot^2 x}{\cot x} = \tan x + \cot x$$

$$h'(x) = \underline{\hspace{1cm} ? \hspace{1cm}}$$

10) $h(x) = \sin 2x$

Hint: $\sin 2x = 2 \sin x \cdot \cos x$

a) $D_x(2 \sin x) = \underline{\hspace{1cm}}?$

b) $D_x(\cos x) = \underline{\hspace{1cm}}?$

c) $h'(x) = \underline{\hspace{1cm}}?$ Use the product Rule for Derivative

